Software Requirements Specification

for

SWEEP - Smart Waste Exchange & Eco Platform

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Chapter 1

Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of Sweep, a digital marketplace for recyclable waste. This document serves as a formal reference for developers, designers, testers, project managers, and other stakeholders throughout the software development lifecycle. It establishes a clear understanding of the system's objectives, scope, features, constraints, and expected behavior, ensuring consistency in design, implementation, testing, and maintenance. Additionally, the system supports environmental sustainability by providing analytics and reports related to waste reduction and carbon impact, promoting informed decision-making and encouraging sustainable waste management practices.

1.2 Intended Audience

This Software Requirements Specification (SRS) is intended for all stakeholders involved in the planning, development, testing, deployment, and maintenance of the Sweep platform. It serves as the primary reference document to ensure every stakeholder has a clear understanding of the system's objectives, requirements, and expected behavior.

Audience	Detailed Description
Product Managers	Use the SRS to understand the business goals, business strategy, product quality expectations, and overall project vision. Define the project scope, prioritize features, assign tasks to team members, and manage the product roadmap.
Software Engineers (Developers)	Use the SRS as a guide for developing the web platform and mobile applications according to the specified functional and non-functional requirements. Ensure that the implementation follows the defined system architecture, business logic, and technical specifications.
Quality Assurance (QA) Engineers	Prepare test plans, test cases, and testing strategies. Verify that all implemented features meet the specified functional and non-functional requirements and satisfy the defined acceptance criteria.
UI/UX Designers	Understand user roles, user journeys, and system workflows to design intuitive, responsive, accessible, and user-friendly interfaces for both the web platform and mobile applications across different operating systems.
Project Managers	Estimate project cost and development timeline, allocate resources efficiently, optimize project expenses, recruit and manage human resources, monitor project progress, and identify and mitigate project risks.

Audience	Detailed Description
Business Stakeholders	Ensure that the platform aligns with business objectives, market needs, and sustainability goals while delivering value to households, garbage collectors, recycling companies, and other stakeholders.
Compliance & Legal Teams	Verify that the platform complies with environmental regulations, data privacy laws, payment regulations, cybersecurity standards, and other applicable legal and regulatory requirements.
System Administrators	Understand the deployment architecture, server configuration, monitoring, backup and recovery procedures, system updates, security management, and maintenance requirements to ensure reliable system operation.
Future Maintenance Teams	Use the SRS as a reference for troubleshooting, debugging, maintaining the system, implementing updates, and developing future feature enhancements without affecting existing functionality.

1.3 Intended Use

This Software Requirements Specification (SRS) is intended for all stakeholders involved in the development and management of the **Sweep** platform. It provides a common reference to ensure that everyone has a shared understanding of the system's requirements and expected functionality.

- **Developers** will use this document to implement the system according to the specified functional and non-functional requirements.
- **Testers and Quality Assurance (QA) Engineers** will use it to design test cases and verify that the system meets the documented requirements.
- **Project Managers** will use it to plan, monitor, and coordinate project activities while ensuring that development aligns with the agreed requirements.
- **UI/UX Designers** will refer to it when designing interfaces that satisfy user and system requirements.
- **Stakeholders**, including clients, supervisors, and leadership teams, will use this document to review, validate, and approve the system requirements before and during development.

1.4 Product Scope

The **Household Waste Recycling Marketplace (Sweep)** is a web-based platform designed to streamline the collection, management, and trading of recyclable household waste by connecting households, waste collectors, recycling companies, and platform administrators within a unified digital ecosystem.

The system addresses the challenges of fragmented waste collection and inefficient recycling by providing a centralized marketplace where recyclable materials can be listed, verified, collected, and sold. By integrating all major stakeholders into a single platform, the system aims to promote sustainable waste management practices while supporting the development of a circular economy.

The platform enables households to create waste listings, schedule pickups or drop-offs, track collection status, receive payments through a digital wallet, and earn rewards for responsible recycling. Waste collectors can manage pickup requests, optimize collection routes, aggregate waste from multiple sources, and monitor their earnings. Recycling companies can subscribe to the platform, purchase verified waste in bulk, monitor available inventory, and analyze supply trends.

Platform administrators oversee user management, transaction monitoring, fraud detection, and system-wide analytics to ensure efficient and transparent operations.

In addition to its core marketplace functionality, the system incorporates features such as AI-assisted waste classification, QR code-based verification, digital payment processing, live pickup tracking, and reward mechanisms to enhance user experience and improve operational efficiency.

Overall, the platform seeks to simplify the recycling process, encourage household participation, optimize waste collection logistics, and create a reliable digital marketplace that benefits both the recycling industry and the environment.

1.5 Risk Definition

The **Household Waste Recycling Marketplace** is a multi-stakeholder platform involving households, waste collectors, recycling companies, payment services, and system administrators. Due to the interaction between software services and real-world waste collection operations, various risks may affect the successful development, deployment, and operation of the system.

In this SRS, a **risk** is defined as any uncertain event or condition that may negatively impact the system's functionality, security, performance, reliability, user experience, or business objectives.

To ensure a consistent understanding of potential challenges, risks within the system are categorized as follows:

- **Technical Risk:** Risks arising from software defects, database failures, third-party API integration issues, network outages, or system performance problems that may affect the platform's functionality.
- **Operational Risk:** Risks associated with the day-to-day operation of the platform, including failures in waste verification, inaccurate data entry, inefficient workflow management, or disruption of platform services.
- **Logistics Risk:** Risks related to the physical collection and transportation of recyclable waste, such as missed pickups, route inefficiencies, collector unavailability, traffic conditions, or delays in transferring waste to aggregation hubs or recycling companies.
- **Security and Privacy Risk:** Risks involving unauthorized access, data breaches, fraudulent activities, identity theft, or misuse of sensitive user information and financial transactions.
- **Financial Risk:** Risks affecting monetary transactions, including payment gateway failures, incorrect payment processing, subscription billing issues, fluctuations in recyclable material prices, or disputes regarding transaction values.
- **Data Integrity Risk:** Risks resulting from inaccurate, incomplete, duplicated, or manipulated data, including incorrect waste classification, false weight reporting, duplicate listings, or inconsistent transaction records.
- **User Adoption Risk:** Risks related to insufficient participation by households, collectors, or recycling companies, as well as resistance to adopting digital waste management practices within the local context of Bangladesh.
- **Regulatory and Environmental Risk:** Risks arising from changes in environmental regulations, waste management policies, legal compliance requirements, or sustainability standards that may affect the platform's operation.

For the purpose of risk evaluation, each identified risk is assessed using two criteria:

- **Probability:** The likelihood of the risk occurring.
- **Impact:** The severity of the consequences if the risk occurs.

Both criteria are classified as **Low**, **Medium**, or **High** to support consistent risk assessment and prioritization throughout the software development lifecycle.

Appropriate risk mitigation strategies should be established to reduce the likelihood or impact of identified risks, while contingency plans should define the actions to be taken if a risk materializes. Note

Chapter 2

Overall Description

2.1 User Classes and Characteristics

This section identifies the four distinct user classes that interact with SWEEP, along with their defining characteristics, technical proficiency, access levels, and verification requirements.

2.1.1 Household User

Attribute	Description
Definition	Individual or family unit that generates recyclable household waste and wishes to sell it.
Technical Proficiency	Low to moderate; primary interaction via mobile app.
Frequency of Use	Occasional to regular (weekly or bi-weekly).
Access Level	Restricted — can create listings, schedule pickups, manage wallet, and review counterparties.
Language	Bengali (primary), English (secondary).
Device	Smartphone (Android primary).
Location	Urban and peri-urban residential areas.
Age Range	18 – 60 years.
Verification	Phone number and email verification required; no KYC.

2.1.2 Collector

Attribute	Description
Definition	Individual or small business that collects waste from households, aggregates it, and resells to recycling companies.
Technical Proficiency	Low to moderate; uses mobile app in the field.
Frequency of Use	Daily operational use.
Access Level	Restricted — can accept pickups, manage routes, track collections, and list bulk inventory.
Language	Bengali (primary).
Device	Smartphone (Android primary).
Location	Operates within assigned zones/wards.
Age Range	20 – 55 years.
Verification	Full KYC[1] required — NID, photo, vehicle registration; admin-approved before activation.

2.1.3 Recycling Company

Attribute	Description
Definition	Registered business entity that purchases recyclable materials from households and collectors for processing.
Technical Proficiency	Moderate to high; uses both web and mobile interfaces.
Frequency of Use	Daily — browsing, purchasing, and reporting.
Access Level	Subscription-gated — full marketplace access, purchase orders, and analytics unlocked by active plan.
Language	Bengali and/or English.
Device	Desktop/web (primary), mobile (secondary).
Location	Industrial or commercial zones.
Age Range	Business account; individual operators typically 25 – 55 years.
Verification	Trade licence, company registration, and e-waste handling licence where applicable; admin-reviewed.

2.1.4 Platform Administrator

Attribute	Description
Definition	Internal SWEEP staff responsible for platform operations, user management, compliance, and configuration.
Technical Proficiency	High; uses admin dashboard via web browser.
Frequency of Use	Daily operational management.
Access Level	Full — all platform data, user controls, financial records, configuration, and audit logs.
Language	English (primary).
Device	Desktop/web.
Location	Internal (office or remote).
Age Range	N/A (internal staff role).
Verification	Secure internal credentials; multi-factor authentication enforced.

2.2 User Needs

This section captures the core needs of each user class — what they must be able to accomplish on SWEEP for it to deliver value to them.

2.2.1 Household User Needs

1. **Account & Identity**
   The household user needs to register with a phone number and email, select their role, and verify their account so that they can securely access the platform.
2. **Waste Listing**
   The household user needs to create waste listings with category, quantity, photos, and an expected price so that buyers can discover and purchase their recyclable materials.
3. **Pickup Scheduling**
   The household user needs to select a convenient date and time window for collection so that a collector arrives without disruption to their routine.
4. **Price Guidance**
   The household user needs to see suggested market prices per waste category so that they can set competitive and fair listing prices.
5. **Earnings & Wallet**
   The household user needs to track earnings, view transaction history, and withdraw funds to mobile banking so that they benefit financially from recycling.
6. **Status Visibility**
   The household user needs real-time updates on listing status, pickup progress, and payment settlement so that they are never left uncertain about the state of a deal.
7. **Trust & Safety**
   The household user needs to rate and review collectors and companies after completed transactions so that the platform maintains service quality.

8. **Accessibility**
   The household user needs the app to function in Bengali and work reliably on low-end Android devices with limited connectivity so that it is accessible to all demographics.

2.2.2 Collector Needs

1. **Onboarding & Verification**
   The collector needs to submit identity and vehicle documents and receive admin approval so that they can legally and legitimately operate on the platform.
2. **Zone & Capacity Control**
   The collector needs to define their service zones and set daily capacity so that they only receive pickup requests they can realistically fulfil.
3. **Pickup Queue Management**
   The collector needs to view, accept, and decline incoming pickup requests with reasons so that they can plan their workday efficiently.
4. **Route Efficiency**
   The collector needs an optimised daily route across accepted pickups so that they minimise travel time and fuel cost.
5. **Collection Verification**
   The collector needs to record actual collected weight, capture proof photos, and update pickup status in real time so that all parties have an accurate record of what was collected.
6. **Bulk Inventory & Resale**
   The collector needs to aggregate pickups into category lots and list them to the marketplace so that they can sell at scale to recycling companies.
7. **Earnings Visibility**
   The collector needs a dashboard showing revenue, platform commissions, and net payout per period and zone so that they can manage margins effectively.

2.2.3 Recycling Company Needs

1. **Subscription Access**
   The recycling company needs to subscribe to a plan and manage renewals so that they maintain uninterrupted access to the marketplace.
2. **Supply Discovery**
   The recycling company needs to browse, search, and filter waste listings by category, quantity, price, distance, and seller rating so that they can efficiently source the materials they need.
3. **Demand Broadcasting**
   The recycling company needs to post material demand requests with category, quantity, price, and location radius so that matching suppliers are automatically notified.
4. **Order & Payment Management**
   The recycling company needs to issue purchase orders, trigger escrow payment holds, and receive release confirmation on delivery so that transactions are secure end-to-end.
5. **Recurring Supply**
   The recycling company needs to set up standing inventory requests on a defined cadence so that their supply chain remains consistent without manual effort each cycle.
6. **Procurement Reporting**
   The recycling company needs dashboards showing purchase volume, spend, supplier mix, and environmental impact so that they can meet procurement targets and sustainability obligations.

2.2.4 Platform Administrator Needs

1. **User Lifecycle Management**
   The administrator needs to search, view, approve, suspend, ban, and reinstate any user account with a required reason so that platform integrity and compliance are maintained.
2. **KYC Approval**
   The administrator needs a review queue for collector and company verification documents so that only legitimate, verified parties operate on the platform.
3. **Fraud Control**
   The administrator needs automated fraud flags and a review queue for suspicious transactions so that weight fraud, fake listings, and payment abuse are detected and acted on promptly.
4. **Pricing & Commission Configuration**
   The administrator needs to set and schedule base rate cards and commission percentages by category and transaction type so that platform economics reflect current market conditions.
5. **Subscription Oversight**
   The administrator needs to create, edit, and archive subscription plans and manage company access so that billing remains accurate and current.
6. **Audit & Compliance**
   The administrator needs immutable, searchable audit logs for all sensitive actions — with export capability — so that the platform can demonstrate compliance to regulators and internal governance.
7. **Operational Reporting**
   The administrator needs platform-wide analytics on collection volume, revenue by stream, and environmental impact so that business performance can be monitored and reported to stakeholders.

2.3 Operating Environment

The **Sweep** platform is designed as a cloud-based web and mobile application supporting waste collection, recycling management, logistics coordination, and marketplace operations.

Environment Component	Detailed Specification
Mobile Application	Android 8.0 or later and iOS 14 or later, with responsive support for smartphones and tablets.
Web Application	Supports Google Chrome 100+, Mozilla Firefox 100+, Safari 15+, and Microsoft Edge 100+.
Backend Server	Cloud-hosted using Amazon Web Services (AWS) or Google Cloud Platform (GCP), with containerized microservices and RESTful APIs.
Database	PostgreSQL for transactional data and a NoSQL database for analytics, logging, and reporting.
Payment Gateway	Integration with bKash, Nagad, Rocket, and bank transfer services for secure digital payments.
Maps & Route Optimization	Google Maps Platform for geolocation, route optimization, navigation, and distance calculation.
SMS Gateway	Integration with local SMS gateway providers such as SSL Wireless for OTP verification and notification services.

Environment Component	Detailed Specification
Push Notifications	Firebase Cloud Messaging (FCM) for real-time alerts and notifications.
Internet Connectivity	Optimized for 3G, 4G, and 5G networks, with graceful degradation under low-bandwidth conditions.
Localization	Supports both English and Bengali languages and uses Bangladeshi Taka (BDT) as the default currency.
Security Environment	HTTPS communication using TLS 1.2 or later, secure password hashing, authentication, authorization, and role-based access control (RBAC).
Hardware Requirements	Mobile devices with at least 2 GB RAM and 100 MB of available storage.
Availability	Designed for 24/7 operation with a target system uptime of 99.9%.
Scalability	Supports horizontal scaling to accommodate increasing numbers of users, transactions, and services.
Backup & Disaster Recovery	Automated daily backups, database replication, and disaster recovery procedures to ensure business continuity.

2.4 Constraints

The Sweep platform must operate within several technical, legal, financial, and operational constraints. These constraints define the limitations and mandatory conditions that must be considered during system design, development, deployment, and maintenance.

- **Regulatory Compliance:** Sweep shall comply with Bangladesh's Digital Security Act, 2018, environmental regulations, and applicable waste management policies. User data must be handled securely according to legal requirements. Collection and trading of regulated waste categories, especially e-waste, shall only be permitted for authorized users and recycling companies with valid environmental licences. System logs, transactions, and activity records shall be maintained for regulatory review when required.
- **Financial Regulations:** All financial transactions performed through Sweep shall comply with Bangladesh Bank regulations. Transactions exceeding regulatory limits shall require proper Know Your Customer (KYC) verification. Payments to households, collectors, and recycling companies shall only be processed through approved payment channels such as bKash, Nagad, and other authorized financial services.
- **E-Waste Licensing:** The platform shall restrict e-waste transactions to authorized parties with valid environmental handling licences. During recycling company onboarding, licence documents shall be verified before enabling e-waste purchasing activities. The system shall implement category-based filtering to distinguish e-waste from other recyclable materials and enforce appropriate access control.
- **Data Residency:** User information, including personal identification documents, financial records, and transaction history, shall be stored and managed according to Bangladesh's data sovereignty requirements. Cloud infrastructure and third-party services shall be selected carefully to ensure compliance with applicable data protection regulations.
- **Device Constraints:** The mobile application shall provide acceptable performance on low-end smartphones with a minimum specification of 2 GB RAM and 16 GB storage. The application shall be optimized to reduce battery consumption, minimize data usage, and provide smooth operation for households, waste collectors, and recycling companies. Images and resources shall be optimized, and unnecessary animations shall be avoided.

- **Payment Gateway Dependency:** Sweep depends on external payment gateway services such as bKash and Nagad for processing digital transactions. If payment gateways become unavailable, the system shall temporarily suspend transactions and clearly notify users. Failed transactions shall be properly logged, monitored, and retried when appropriate to prevent financial loss or data inconsistency.

2.5 Assumptions

ASMP-01: Smartphone & Internet Access

Assumption: All users have a smartphone with at least 3G internet connectivity from all places of the country.

Per User Perspective:

- **Households:** Assumed to own at least a basic Android smartphone with minimal specifications. No feature-phone or offline mode is planned.
- **Collectors:** Assumed to have continuous internet connectivity in the field for map updates, QR code scanning, and real-time order updates.
- **Recycling Companies:** Assumed to have reliable office internet or Wi-Fi for dashboard access and bulk order management.

Risk if Wrong: Rural households or collectors with only 2G connectivity may experience degraded performance or be unable to use important platform features.

ASMP-02: bKash/Nagad Account Ownership

Assumption: Collectors have an active bKash or Nagad account. Households have at least a registered mobile phone number for SMS verification.

Per User Perspective:

- **Households:** OTP-based verification requires only a valid and active mobile phone number.
- **Collectors:** Must already possess a working Mobile Financial Service (MFS) account. Sweep does not assist in creating one.

Risk if Wrong: Collectors without a valid bKash or Nagad account will not be able to receive payouts and therefore cannot participate on the platform.

ASMP-03: Recycling Companies Are Registered Legal Entities

Assumption: All recycling companies onboarded onto the platform are legally registered and possess valid trade licenses and supporting documentation.

Per User Perspective:

- **Recycling Companies:** Assumed to be legitimate organizations with valid legal documents rather than informal scrap dealers.
- **Admins:** Assume submitted KYC documents are authentic. The platform performs verification but not forensic validation.

Risk if Wrong: Fraudulent organizations using fake licenses could manipulate the marketplace, particularly for high-value recyclable materials such as e-waste.

ASMP-04: Pricing Engine Data Availability

Assumption: The dynamic pricing engine will obtain sufficient historical transaction data within the first 90 days after launch to generate meaningful price recommendations. Until then, administrator-defined base rate cards will be used.

Per User Perspective:

- **Households:** Initially receive administrator-defined prices rather than market-driven recommendations.
- **Collectors:** Quote prices based on predefined rates instead of dynamic market demand.
- **Recycling Companies:** Must rely more on manual price negotiation during the early operational period.
- **Admins:** Responsible for configuring and maintaining base prices for different waste categories until sufficient transaction data is collected.

Risk if Wrong: If 90 days is insufficient to collect adequate transaction data, price recommendations may remain inaccurate for a longer period.

ASMP-05: Google Maps API Availability and Budget

Assumption: Google Maps Platform APIs will remain available and financially sustainable for the projected transaction volume during the first 12 months of operation.

Per User Perspective:

- **Collectors:** Depend on Google Maps for routing, zone management, and navigation to pickup locations.
- **Households:** Depend on Maps API for address geocoding during registration and waste listing.
- **Recycling Companies:** Use distance calculations to optimize sourcing and logistics.

Risk if Wrong: Unexpected increases in API costs may require limiting map-related functionality or migrating to alternatives such as OpenStreetMap.

ASMP-06: bKash/Nagad API Reliability

Assumption: The APIs provided by bKash and Nagad are capable of supporting the required transaction volume and reliable webhook delivery for Sweep's escrow and payout processes.

Per User Perspective:

- **Households:** Expect payments to be processed promptly after delivery confirmation.
- **Collectors:** Depend on timely escrow release triggered by QR-code confirmation.
- **Recycling Companies:** Require dependable payment processing for high-volume bulk transactions.

Risk if Wrong: Frequent webhook failures could require manual transaction reconciliation, increasing the operational workload for administrators.

ASMP-07: Admin Domain Knowledge

Assumption: Platform administrators are trained Sweep personnel with sufficient knowledge of the waste recycling industry to perform KYC verification, fraud detection, and dispute resolution accurately.

Per User Perspective:

- **Collectors:** Depend on administrators for fair account approval and zone assignment.
- **Recycling Companies:** Depend on administrators to correctly verify trade licenses and e-waste permits.
- **Households:** Rely on administrators to investigate fraud reports and resolve disputes appropriately.

Risk if Wrong: Insufficiently trained administrators may approve fraudulent companies, reject legitimate collectors, or mishandle disputes, reducing trust in the platform.

Chapter 3

Requirements

3.1 Functional Requirements

3.1.1 FR-01 — Registration

Covers multi-role account creation for Household users, Local Collectors, Global Collectors, and Recycling Companies.

FR-1.1 Household Registration

User Story
"As a new household user, I want to register an account with my personal and address details, so that I can request waste pickups and sell recyclable materials."

Details

- Required fields: full name, phone number, email address, NID number, home address (with map pin for pickup location), password.
- A verification link is sent to the registered email upon successful submission.
- Business rule: one account per verified NID number and per verified phone number.
- Business rule: if an NID or email belongs to a previously suspended or banned account, registration is permanently blocked for that NID and email.
- Validation: email must be a valid format; phone must match Bangladesh format (+880); password minimum 8 characters with at least 1 number and 1 symbol; NID must be 10 or 17 digits.
- The pickup address is pinned on a map and saved as the default pickup location.

Acceptance Criteria

- Given a new user submits valid details, then an unverified account is created and a verification link is sent.
- Given a phone or NID already registered, then registration is blocked with a clear message.
- Given a suspended NID or email, then registration is permanently blocked.
- Given an invalid email or weak password, then field-level errors are shown and submission is prevented.

Failure Cases & Edge Cases

- Duplicate NID or phone: blocked with "Account already exists" message.
- Suspended NID or email: blocked with "This identity is not eligible for registration" message.
- Network failure during submission: user is prompted to retry.
- Map pin outside service coverage area: a warning is shown that no collectors are currently available in that area.

FR-1.2 Local Collector Registration

User Story
"As a new local collector, I want to register with my identity and vehicle documents, so that I can accept household waste pickup requests in my area."

Details

- Required fields: full name, phone number, email, NID number, service area/zone selection, password.
- Optional but recommended: profile photo.
- Local collectors collect waste directly from households, categorize it, and hold it until a global collector picks it up.
- Account status is set to Pending KYC after submission; the collector cannot accept pickups until admin approves.
- Business rule: NID and email must not belong to any previously suspended or banned account.
- Validation: same rules as household registration.

Acceptance Criteria

- Given valid details are submitted, then a Pending KYC account is created and the collector is notified to await approval.
- Given a duplicate NID or suspended identity, then registration is blocked.
- Given admin approves the KYC, then the collector gains access to the pickup request system.
- Given admin rejects the KYC, then the collector is notified with a reason and may resubmit.

Failure Cases & Edge Cases

- Blurry or unreadable document: admin rejects with reason; resubmission is allowed.
- Collector tries to accept pickups before KYC approval: action is blocked with a "Pending Approval" message.

FR-1.3 Global Collector Registration

User Story
"As a new global collector (truck driver), I want to register with my driving licence and vehicle details, so that I can collect bulk waste from local collectors and deliver it to recycling companies."

Details

- Required fields: full name, phone number, email, NID number, driving licence number, vehicle registration number, vehicle type and capacity (kg), service regions, password.
- Global collectors pick up large, aggregated bulk lots from local collectors and transport them to recycling companies.
- Account status is Pending KYC until admin approves.
- Business rule: NID, driving licence, and email must not belong to any previously suspended or banned account.
- Validation: driving licence format validated; vehicle capacity must be a positive number.

Acceptance Criteria

- Given valid details, then a Pending KYC account is created.
- Given admin approves, then the global collector can view and accept bulk lot pickup requests from local collectors.
- Given admin rejects, then the global collector is notified with a reason and may resubmit corrected documents.

Failure Cases & Edge Cases

- Duplicate driving licence: registration is blocked.
- Vehicle capacity entered as zero or negative: validation error shown.

FR-1.4 Recycling Company Registration

User Story
"As a recycling company representative, I want to register a company account, so that I can subscribe to a plan, browse waste lots, and place purchase orders."

Details

- Required fields: company name, trade licence number, contact person name, phone number, email, company address, waste categories handled (e.g., Plastic, Metal, E-waste), password.
- Company account is Pending KYC until admin verifies the trade licence.
- After KYC approval, the company must purchase a subscription plan before accessing the marketplace newsfeed and placing orders.
- Business rule: trade licence number and email must be unique and must not belong to a suspended entity.

Acceptance Criteria

- Given valid registration details, then a Pending KYC account is created.
- Given admin approves the trade licence, then the company is prompted to select a subscription plan.
- Given the subscription is purchased, then the company gains access to the newsfeed and dashboard.
- Given a duplicate trade licence, then registration is blocked.

Failure Cases & Edge Cases

- Company tries to access the marketplace before subscribing: redirected to the subscription selection screen.
- Trade licence expired: admin may flag for follow-up but registration proceeds; access is reviewed periodically.

3.1.2 FR-02 — Login

Covers secure authentication for all user roles.

User Story
"As a registered user, I want to log in with my phone number or email and password, so that I can securely access my role-specific dashboard."

Details

- Login identifier: phone number or email address.
- Upon successful login, the system routes the user to their role-specific dashboard (Household, Local Collector, Global Collector, or Company).
- "Remember me" option keeps the session active for 30 days.
- Business rule: unverified accounts cannot log in; a prompt to verify is shown.
- Business rule: suspended or banned accounts are blocked at login with a support contact message.
- Security rule: after 5 consecutive failed login attempts, the account is temporarily locked for 15 minutes.
- All sessions are invalidated on password reset or account suspension.

Acceptance Criteria

- Given valid credentials, then the user is routed to their role-based dashboard.
- Given 5 consecutive failed attempts, then the account is temporarily locked for 15 minutes.
- Given an unverified account, then the user is prompted to complete email verification.
- Given a suspended account, then login is blocked and a support message is shown.

Failure Cases & Edge Cases

- Wrong credentials: a generic "Invalid credentials" message is shown (no enumeration of which field is wrong).
- Account locked: countdown timer shown; user may request an unlock email.
- Pending KYC collector tries to log in: login succeeds but access to pickup functions is restricted until KYC is approved.

3.1.3 FR-03 — Household Waste Listing

Covers waste categorization, listing creation, photo upload, and pickup scheduling by household users.

User Story
"As a household user, I want to create a waste listing with category, weight, photos, and a preferred pickup time, so that a local collector can come and collect it."

Details

- The household selects a waste category from: Plastic, Paper, Metal, Glass, E-waste, Organic, Textile, or Mixed.
- Sub-categories are available where relevant (e.g., PET vs. HDPE for plastic).
- The household enters estimated weight or quantity and an expected price per kg.

- The system displays a suggested price range based on the selected category and region.
- Up to 5 photos of the waste can be attached (max 5 MB each; JPEG, PNG, or WebP).
- The pickup location defaults to the saved home address; the household may select an alternate saved address.
- The household selects an available pickup date and time window.
- Once submitted, the listing enters the system and a nearby local collector receives a notification.
- The household can view the status of their listing: Pending, Accepted, Collected, or Cancelled.
- Cancellation is free up to 2 hours before the scheduled pickup time; late cancellations are flagged.
- Business rule: at least one photo and one category are required to submit a listing.
- Business rule: price set far above the suggested range triggers a soft warning but does not block submission.

Acceptance Criteria

- Given valid listing details and at least one photo, then the listing is created and a nearby local collector is notified.
- Given a missing photo or category, then submission is blocked with field-level errors.
- Given a household cancels within the free window, then the listing is cancelled without penalty.
- Given a household cancels after the free window, then a late-cancel flag is recorded on the account.
- Given a collector accepts the listing, then the household receives a confirmation notification with the collector's details.

Failure Cases & Edge Cases

- Oversized photo: the system attempts compression; if still too large, the file is rejected with guidance.
- No local collector available in the area: the household is notified and the listing remains open for 24 hours before auto-expiry.
- Household submits duplicate listing for same address and same day: a confirmation prompt is shown before submission.
- E-waste category selected: listing is routed only to collectors certified for e-waste handling.

3.1.4 FR-04 — Household Wallet

Covers the household user's digital wallet, earnings tracking, transaction history, and withdrawal to mobile banking.

User Story
"As a household user, I want a digital wallet that shows my earnings from waste sales, so that I can track my income and withdraw money to my mobile banking account."

Details

- Every household account has an automatically created digital wallet with an initial balance of zero.
- Wallet balance is shown in two states: Pending (funds awaiting confirmation) and Available (cleared funds ready for withdrawal).

- Funds move from Pending to Available once the collector confirms pickup completion.
- A full ledger view shows every credit and debit with date, transaction ID, and counterparty.
- Earnings are shown broken down by day, week, and month, and further by waste category.
- Earnings reports can be exported as CSV or PDF.
- Withdrawals are supported to bKash, Nagad, and registered bank accounts.
- A minimum withdrawal threshold applies; withdrawals below the threshold are blocked.
- A verified payout account (mobile banking number or bank account) must be linked before withdrawal.
- The household can rate and review the local collector after each completed transaction (1–5 stars with an optional comment).
- One review is permitted per completed transaction.

Acceptance Criteria

- Given a completed pickup, then funds move from Pending to Available in the wallet.
- Given an available balance above the minimum threshold, then a withdrawal is initiated successfully.
- Given no payout account is linked, then withdrawal is blocked and the user is prompted to add one.
- Given a completed transaction, then the household can submit a rating and it updates the collector's average score.

Failure Cases & Edge Cases

- Withdrawal below minimum threshold: blocked with threshold amount shown.
- Gateway failure during withdrawal: funds are not deducted; user is notified and prompted to retry.
- Duplicate withdrawal request: second request is blocked; idempotency enforced.
- Household tries to review same transaction twice: second review is blocked.

3.1.5 FR-05 — Collector's Pickup Management

Covers local collector zone assignment, pickup request handling, collection verification, and bulk lot aggregation for global collector handover.

User Story
"As a local collector, I want to manage incoming pickup requests from households in my area, record collected waste, and prepare aggregated bulk lots for global collectors to pick up."

Details

- After KYC approval, the local collector defines their service zones/wards on a map and sets a daily pickup capacity.
- The local collector receives real-time notifications when a household in their zone submits a pickup request.
- The collector can Accept or Decline each request; declined requests show a reason.
- Accepted requests appear in the collector's daily schedule.
- During or after a pickup, the collector records the actual collected weight, uploads proof photos, and updates the pickup status (En Route → Arrived → Collected → Completed).

- Business rule: actual weight must be entered before a pickup can be marked Completed.
- Business rule: if the recorded weight differs significantly from the listed weight, the pickup is flagged for review.
- The local collector aggregates completed pickups by waste category into bulk lots.
- A bulk lot shows total weight per category and is marked as Ready for Global Collector.
- The local collector's dashboard shows all waste currently held (category, weight, and status: held / ready for pickup by global collector).
- The local collector cannot accept new requests beyond their set daily capacity.

Acceptance Criteria

- Given an incoming request, then the collector receives a notification and can accept or decline.
- Given the collector accepts, then the household is notified and the slot is confirmed.
- Given the collector marks a pickup Completed with weight and photo, then the household's payment moves to Available.
- Given a weight variance beyond tolerance, then the pickup is flagged for admin review.
- Given pickups are aggregated into a bulk lot, then the lot appears in the collector's waste dashboard as Ready for Global Collector.

Failure Cases & Edge Cases

- Collector tries to accept beyond daily capacity: blocked with a capacity-full message.
- Collector goes offline mid-route: pickup status remains unchanged until the collector reconnects and updates it.
- Pickup request expires with no response (24 hours): the request is auto-cancelled and the household is notified.
- Mixed-category bulk lot attempted: system enforces one category per lot.

3.1.6 FR-06 — Waste Dashboard for Collector

Covers the local collector's dashboard showing held waste inventory, earnings summary, and payout status.

User Story
"As a local collector, I want a dashboard that shows all the waste I currently hold, my total earnings, and any pending payments, so that I can track my inventory and finances."

Details

- The dashboard displays a summary of all waste currently held by the collector, broken down by category and total weight.
- Each lot entry shows: waste category, total weight (kg), date collected, and current status (Held / Ready for Global Collector / Handed Over).
- The earnings section shows: total revenue earned, platform commission deducted, and net payout.
- Earnings can be filtered by day, week, month, and by waste category.
- The dashboard shows any payments that are pending (awaiting global collector confirmation) vs. available (cleared).

- The collector can initiate a withdrawal from available balance to bKash, Nagad, or bank account.
- A transaction history ledger shows all past credits and debits.
- Earnings reports can be exported as CSV or PDF.

Acceptance Criteria

- Given the collector opens the dashboard, then held waste by category and weight is displayed accurately.
- Given a global collector completes a handover, then the relevant lot's status updates to Handed Over and payment moves to Available.
- Given the collector applies a period filter, then earnings figures update accordingly.
- Given available balance is above the minimum threshold, then withdrawal is initiated successfully.

Failure Cases & Edge Cases

- Dashboard loads with no data: a friendly empty state is shown.
- Withdrawal below minimum threshold: blocked with guidance.
- Global collector handover not yet confirmed: payment remains in Pending state; dashboard reflects this clearly.

3.1.7 FR-07 — Subscription Management of Companies

Covers recycling company subscription plan selection, renewal, and access control.

User Story
"As a recycling company, I want to subscribe to a plan after KYC approval, so that I can access the waste marketplace newsfeed and place purchase orders."

Details

- After KYC approval, the company is presented with available subscription plans (e.g., Basic, Pro, Enterprise) with features and pricing.
- Payment is made via bKash, Nagad, or bank transfer.
- The subscription is time-bound (monthly or annual); an invoice is generated on each renewal.
- Business rule: marketplace access (newsfeed and order placement) is gated behind an active subscription.
- Business rule: if a renewal payment fails, a grace period of 3 days is granted; after that, marketplace access is restricted.
- The company can upgrade or downgrade the plan; changes take effect at the next renewal cycle.
- Admins can create, edit, archive, and comp (grant free access to) subscription plans.
- Archiving a plan that currently has active subscribers is blocked; the plan is marked as no-longer-available for new signups instead.

Acceptance Criteria

- Given KYC approval, then the company is prompted to select a subscription plan before accessing the marketplace.
- Given a successful payment, then marketplace access is unlocked and an invoice is generated.

- Given a failed renewal past the grace period, then marketplace access is restricted and a renew prompt is shown on login.
- Given an admin archives an in-use plan, then it is marked unavailable for new signups but existing subscribers are unaffected.

Failure Cases & Edge Cases

- Payment gateway timeout: transaction is shown as pending; balance is not deducted; user is prompted to retry.
- Company tries to access marketplace before subscribing: redirected to subscription selection screen.
- Duplicate payment request: second request is blocked; idempotency enforced.

3.1.8 FR-08 — Company Dashboard

Covers the two primary screens available to subscribed recycling companies: the waste lot newsfeed and the procurement dashboard.

FR-8.1 Waste Lot Newsfeed

User Story
"As a subscribed recycling company, I want to browse a region-based newsfeed of available waste lots held by local collectors, so that I can add lots to my list and place purchase orders."

Details

- The newsfeed displays available waste lots from local collectors, filtered by the company's registered service regions.
- Each lot card shows: waste category, total weight (kg), collector name and rating, region, and asking price per kg.
- The company can filter lots by category, weight range, price range, region, and collector rating.
- The company can add a lot to their Order List for later, or place an order immediately.
- Placed orders are sent to the local collector for confirmation; a global collector is then assigned for transport.
- Business rule: only active-subscription companies can view lot details and place orders.
- Business rule: E-waste lots are visible only to companies licensed for e-waste handling.

Acceptance Criteria

- Given an active subscription, then available lots in the company's regions are displayed.
- Given the company adds a lot to the Order List, then it is saved and accessible from the dashboard.
- Given the company places an order, then the local collector is notified and a global collector is assigned.
- Given an expired subscription, then lot details are hidden and a renew prompt is shown.

Failure Cases & Edge Cases

- Lot becomes unavailable while being viewed: a "No longer available" message is shown on order attempt.
- No lots available in the company's region: a friendly empty state with suggestions to expand region filter is shown.
- E-waste lot viewed by unlicensed company: the lot is hidden from their feed.

FR-8.2 Company Procurement Dashboard

User Story
"As a subscribed recycling company, I want a dashboard showing my total orders, pending and ongoing orders, and waste type breakdown with financial summary, so that I can track my procurement activity."

Details

- The dashboard displays: total orders placed, pending orders (awaiting collector confirmation), ongoing orders (in transit), and completed orders.
- A waste type breakdown chart shows how much of each category (kg) has been ordered in the current period.
- A financial summary shows: total spend, any pending payments, and transaction history.
- The company can filter the dashboard by date range and waste category.
- Orders can be exported as a CSV or PDF report.
- Environmental impact summary shows estimated CO_2 savings from recycled materials.

Acceptance Criteria

- Given order history exists, then total, pending, ongoing, and completed orders are displayed accurately.
- Given the company applies a category filter, then figures update to show only that category's data.
- Given a completed delivery, then the order moves to Completed and the financial summary updates.

Failure Cases & Edge Cases

- No order history: a friendly empty state is shown.
- Export with an empty date range: blocked with a prompt to select a valid range.

3.1.9 FR-09 — Notification System

Covers real-time, push, and in-app notifications for all platform users, including households, local collectors, global collectors, recycling companies, and administrators.

User Story *"As a platform user, I want to receive real-time notifications about activities relevant to me, so that I can stay informed and take required actions promptly."*

Details

- Households receive notifications for pickup request updates, collector assignment, pickup acceptance, collector arrival, pickup completion, payment availability, withdrawal status, review reminders, account status changes, and admin messages.
- Local collectors receive a push notification and an in-app alert when a household in their registered service zone submits a pickup request.
- The local collector notification includes: household name, waste category, estimated weight, pickup address (zone-level), and preferred pickup time.
- The local collector can Accept or Decline directly from the notification.
- If no response is given within 30 minutes, the pickup request is automatically forwarded to the next available local collector in the zone.

- Local collectors also receive notifications for upcoming pickups, pickup cancellations, weight variance flags, bulk lot status changes, global collector assignments, payments, account status changes, and admin messages.
- Global collectors receive a notification when a local collector marks a bulk lot as Ready for Pickup or when a new bulk lot is assigned to them.
- The global collector notification includes: local collector name, lot categories, total weight, pickup location, destination company, and scheduled pickup information.
- Global collectors also receive notifications for pickup schedule changes, handover updates, delivery reminders, payment updates, account status changes, and admin messages.
- Recycling companies receive notifications for new relevant waste lots, order confirmation, global collector assignment, order transit and delivery updates, payment status, subscription activation, upcoming renewal, failed renewal, grace-period status, KYC or licence status changes, and admin messages.
- Administrators receive notifications for new KYC submissions, fraud flags, significant weight variance, suspicious account activity, payment or payout failures, subscription issues, pickup or delivery delays, security events, audit logging failures, and other operational issues requiring attention.
- Notifications are categorized based on their type, such as Pickup, Order, Payment, Wallet, Subscription, KYC, Security, Fraud, Account, and System.
- Notifications may have Normal, Important, Urgent, or Critical priority depending on the event.
- Notification preferences can be configured per category and available delivery channel (e.g., mute non-urgent updates). Mandatory security and critical account notifications cannot be disabled.
- All notifications are stored in an in-app inbox with timestamp and read/unread status.
- The application displays an unread notification count, and users can mark individual notifications or all notifications as read.
- Where applicable, notifications provide contextual actions such as View Pickup, Accept, Decline, Track Pickup, View Order, View Wallet, Retry Payment, Review KYC, or Renew Subscription.

Acceptance Criteria

- Given a household submits a request in a local collector's zone, then the eligible collector receives a push and in-app notification within 30 seconds.
- Given the local collector accepts from the notification, then the household is notified of acceptance.
- Given no local collector responds within 30 minutes, then the request is forwarded to the next available collector.
- Given a pickup status changes, then the household receives the relevant pickup status notification.
- Given a completed pickup causes the household's payment to become available, then the household receives a payment notification.
- Given a bulk lot is marked Ready, then the assigned global collector receives a pickup notification.
- Given a global collector is assigned to an order, then the relevant local collector and recycling company receive the appropriate assignment notification.
- Given an order moves to In Transit or Delivered, then the recycling company receives the corresponding order status notification.
- Given a recycling company's subscription is approaching renewal or its renewal payment fails, then the company receives the appropriate subscription notification.

- Given a new KYC submission or high-risk fraud flag is created, then the appropriate administrator receives a notification.
- Given a notification is generated, then it appears in the recipient's in-app inbox with unread status.
- Given a user opens or marks a notification as read, then its read/unread status is updated.
- Given a notification contains a contextual action, then selecting the action opens the corresponding record or workflow.

Failure Cases Edge Cases

- Push notification token invalid (user reinstalled app): token is refreshed silently where possible; the notification remains available in the in-app inbox.
- User is offline when a notification is sent: the notification is queued and delivered when the user comes online.
- Push or email delivery fails: the in-app notification remains available and failed delivery is retried where applicable.
- Duplicate notification event is received: duplicate notifications for the same event are prevented.
- All local collectors in a zone decline or fail to respond: the household is notified and the request remains open for 24 hours before auto-expiry.
- A referenced pickup, order, bulk lot, or transaction is cancelled or no longer available: the notification remains in history, but opening it shows the current status or an appropriate unavailable message.
- User disables an optional notification category: the preference is respected, while mandatory security and critical account notifications continue to be delivered.
- Notification service is temporarily unavailable: the originating business operation is not duplicated or rolled back; the notification is queued for retry.
- User is logged in on multiple devices: notification read/unread status remains synchronized across active sessions.

3.1.10 FR-10 — Payment

Covers wallet top-up, payment processing, payout to mobile banking, and commission deduction for all user roles.

User Story
"As a user, I want to make and receive payments through bKash or Nagad, so that all financial transactions on the platform are fast, secure, and traceable."

Details

- Supported payment channels: bKash, Nagad, and registered bank accounts.
- Payment flows: (a) company pays for a waste lot order; (b) platform pays local collector after global collector confirms handover; (c) platform pays household after local collector confirms pickup.
- Platform commission is automatically calculated and deducted from each transaction before payout.
- Commission rates are configurable by the admin per transaction type and category.
- Every transaction generates an immutable ledger entry with: amount, commission deducted, net payout, timestamp, and transaction ID.
- Payment status transitions: Initiated → Processing → Completed / Failed.

- On payment failure, funds are not deducted and the user is notified with a retry option.
- Duplicate payment requests are detected and blocked (idempotency key enforced).
- Payout requires a verified mobile banking number or bank account linked to the user's profile.
- Minimum payout threshold applies; payouts below the threshold are blocked.
- All webhook callbacks from payment gateways are signature-verified before processing.

Acceptance Criteria

- Given a successful payment, then the wallet is credited exactly once and a ledger entry is created.
- Given a duplicate webhook, then no double-credit occurs.
- Given a payment failure, then no balance change occurs and the user is notified.
- Given commission rates are updated by admin, then new transactions use the updated rate; past transactions retain the rate applied at the time.
- Given a payout request above the minimum threshold, then the payout is initiated and status is trackable.

Failure Cases & Edge Cases

- Gateway timeout: transaction shown as Pending; auto-retry after 5 minutes; user notified.
- Payout below minimum threshold: blocked with the threshold amount displayed.
- Unverified payout account: payout blocked; user prompted to link and verify a payout method.
- Concurrent debit requests: locking mechanism prevents overdraft; only one request processes at a time.

3.1.11 FR-11 — Admin Portal

Covers the platform administrator's tools for user management, KYC review, pricing configuration, audit logging, and operational reporting.

User Story
"As a platform administrator, I want a comprehensive admin portal, so that I can manage users, review KYC submissions, configure pricing, monitor platform health, and maintain compliance."

Details

- **User Management:** admin can search, view, approve, suspend, ban, and reinstate any user account. A reason is required for every suspend or ban action. Bans block login immediately.
- **KYC Review:** a dedicated queue shows pending KYC submissions from collectors and companies with document previews. Admin can approve or reject with a written reason.
- **Pricing & Commission Configuration:** admin sets base price ranges per waste category and region, and commission percentages per transaction type. Changes are versioned with an effective date; past transactions retain the rate applied at the time. Overlapping effective dates are blocked.
- **Subscription Management:** admin can create, edit, price, and archive subscription plans. Archiving an in-use plan marks it unavailable for new signups without affecting existing subscribers.
- **Fraud Detection Queue:** rule-based flags (e.g., weight variance, rapid account creation) surface suspicious activity. Admin can clear or escalate flags; all decisions are logged.

- **Audit Logs:** every sensitive admin action (ban, payout override, rate change, KYC decision) is recorded as an immutable, tamper-evident log entry with actor, action, timestamp, and target. Logs can be filtered, searched, and exported for compliance.
- **Operational Reports:** platform-wide analytics showing collection volume by category and region, revenue by stream (commission, subscription), and environmental impact (CO_2 savings). Reports can be exported.

Acceptance Criteria

- Given a flagged user, when admin suspends them with a reason, then they cannot log in and the action is recorded in the audit log.
- Given a pending KYC submission, when admin approves, then the user gains their role's permissions.
- Given a pricing update with a future effective date, when that date arrives, then new transactions use the updated rate.
- Given a sensitive admin action, when performed, then an immutable audit log entry is created.
- Given an admin attempts to delete an in-use subscription plan, then it is archived rather than deleted.

Failure Cases & Edge Cases

- Admin attempts to act on a protected super-admin account: action is blocked.
- Audit log write failure: the action it would have recorded is blocked rather than performed unlogged.
- Overlapping effective dates for pricing: conflict is rejected at save time with a clear error.
- Admin bans a user with an active ongoing order: the order is flagged and placed on hold pending resolution.

3.1.12 FR-12 — Account Security

Covers account suspension enforcement, NID/email blacklisting, password reset, and session security for all user roles.

User Story
"As a household or collector user I want the platform to enforce strict access controls so that suspended or banned identities cannot re-register, and all users can securely reset their passwords and manage their sessions."

Details

- **Suspension Enforcement:** when an account is suspended or banned, the associated NID number and email address are blacklisted. Any future registration attempt using a blacklisted NID or email is permanently blocked regardless of the phone number used.
- **Password Reset:** a user can request a password reset via their registered email. A secure, time-limited reset link (valid for 15 minutes) is sent. Using the link invalidates all active sessions for that account. The new password must differ from the previous password and must meet the strength policy.
- **Session Management:** sessions expire after 30 minutes of inactivity. The "Remember me" session has a maximum lifetime of 30 days. Admins can force-expire all sessions for any account.
- **Role-Based Access Control (RBAC):** each user role (Household, Local Collector, Global Collector, Company, Admin) has a defined set of permissions. Access to any protected endpoint checks role and ownership. Unauthorized access returns a 403 response and is logged.
- **Account Reinstatement:** a suspended account can be reinstated by an admin. On reinstatement, the NID and email are removed from the blacklist and the user can log in again.

- **KYC Blocking:** if a collector or company's KYC is rejected and they do not resubmit within 30 days, the account is automatically deactivated.

Acceptance Criteria

- Given a banned account's NID is used in a new registration, then registration is permanently blocked.
- Given a valid password reset link is clicked within 15 minutes, then all active sessions are invalidated and the user sets a new password.
- Given an expired reset link is opened, then the user is prompted to request a new one.
- Given a household user attempts to access a collector-only endpoint, then access is denied (403) and the attempt is logged.
- Given an admin reinstates a suspended account, then the NID and email are removed from the blacklist and login is restored.

Failure Cases & Edge Cases

- Reset link used after expiry: blocked; user must request a new link.
- Reset link used twice: second use is blocked; link is single-use.
- User changes email after requesting a reset: the reset link sent to the old email is invalidated.
- Collector KYC not resubmitted in 30 days: account is auto-deactivated and the user is notified by email.
- Admin force-expires sessions for an account with an active session: the next API call from that session returns a 401 Unauthorized response and the user is redirected to login.

3.2 Non Functional Requirements

3.2.1 Performance Requirements

NFR-1: Response Time
The system shall respond to standard user requests within 2 seconds under normal operating conditions and within 5 seconds during peak usage.

NFR-2: Concurrent Users
The system shall support a minimum of 5,000 concurrent active users without significant degradation in system performance.

NFR-3: Database Query Performance
The system shall process database retrieval operations within 1 second for standard queries and 3 seconds for complex analytical queries.

NFR-4: Transaction Processing Time
The system shall complete payment and wallet transactions within 3–5 seconds.

NFR-5: Notification Delivery Performance
The system shall deliver push notifications within 5 seconds, and SMS or email notifications within 30 seconds.

NFR-6: Waste Image Processing Time
The system shall classify uploaded waste images and return results within 3–7 seconds.

3.2.2 Reliability Requirements

NFR-7: System Availability
The system shall maintain at least 99.5% availability during operational hours excluding scheduled maintenance periods.

NFR-8: Failure Recovery
The system shall recover from unexpected system failures within 10 minutes.

NFR-9: Data Backup
The system shall automatically create incremental backups every 6 hours and full backups every 24 hours.

NFR-10: Data Recovery
The system shall restore system functionality and recover data within 30 minutes after a critical failure.

3.2.3 Security Requirements

NFR-11: Authentication Security
The system shall require user authentication before granting access to protected resources.

NFR-12: Password Security
User passwords shall be stored using secure hashing algorithms and shall contain:

- Minimum 8 characters
- At least one uppercase letter
- At least one lowercase letter
- At least one numeric value
- At least one special character

NFR-13: Payment Security
The system shall encrypt all payment and financial transaction data using 256-bit encryption.

NFR-14: Session Management
User sessions shall automatically expire after 30 minutes of inactivity.

NFR-15: Login Attempt Restriction
The system shall temporarily lock accounts after 5 consecutive failed login attempts for 15 minutes.

NFR-16: Audit Logging
The system shall retain activity and transaction logs for at least 12 months.

3.2.4 Scalability Requirements

NFR-17: User Scalability
The system architecture shall support future scaling from 5,000 users to 100,000+ users without major redesign.

NFR-18: Database Scalability
The database shall support at least 1 million records of users and transaction data.

NFR-19: Modular Expansion
The system shall support addition of future modules such as:

- Smart waste bins
- AI recommendations
- IoT integration
- Advanced analytics

without major structural changes.

3.2.5 Availability Requirements

NFR-20: Service Availability
The system shall remain accessible 24 hours per day and 7 days per week.

NFR-21: Maintenance Notification
Users shall receive maintenance notifications at least 24 hours before scheduled downtime.

NFR-22: Maximum Downtime
Total scheduled downtime shall not exceed 4 hours per month.

3.2.6 Usability Requirements

NFR-23: Interface Simplicity
The system shall provide a user-friendly interface suitable for users with basic computer knowledge.

NFR-24: Navigation Efficiency
Users shall complete major tasks such as waste listing or payment processing within 3–5 interactions.

NFR-25: Error Message Clarity
The system shall display understandable error messages with suggested corrective actions.

NFR-26: Learning Time
New users shall be able to understand core system operations within 10–15 minutes of use.

3.2.7 Compatibility Requirements

NFR-27: Browser Compatibility
The system shall support the latest versions of:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari

NFR-28: Device Compatibility
The system shall support:

- Desktop devices
- Tablets
- Smartphones

NFR-29: Operating System Compatibility
The system shall support:

- Windows
- Linux
- Android
- iOS

3.2.8 Maintainability Requirements

NFR-30: Modular Design
The system shall follow modular architecture principles to simplify future maintenance.

NFR-31: Code Documentation
At least 90% of critical modules shall contain appropriate documentation and comments.

NFR-32: Error Monitoring
The system shall automatically log and report system errors.

NFR-33: Software Updates
System updates shall not affect previously stored data or active user accounts.

3.2.9 Data Integrity Requirements

NFR-34: Data Consistency
The system shall ensure consistency among all transaction and user data.

NFR-35: Data Validation
The system shall validate all input data before storage.

NFR-36: Duplicate Prevention
The system shall prevent duplicate transaction records and duplicate account registration.

3.2.10 Legal and Compliance Requirements

NFR-37: Data Privacy Compliance
The system shall maintain confidentiality and privacy of user data according to applicable regulations.

NFR-38: Transaction Transparency
The system shall provide complete transaction histories for all financial activities.

NFR-39: User Consent
The system shall obtain user consent before collecting personal information.

3.2.11 Environmental Requirements

NFR-40: Carbon Calculation Accuracy
Environmental impact calculations shall maintain an estimated accuracy of at least 90%.

NFR-41: Report Generation Time
Environmental and carbon reports shall be generated within 5 seconds.

NFR-42: Analytics Refresh Rate
System analytical dashboards shall refresh data every 5 minutes.

Appendices

Appendix A

Glossary